

# CHRISTOPHER E. H. LUTSKO

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## BIOGRAPHICAL

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Born: April 26 1994

Pronouns: He/Him

Citizen: USA/UK

Spoken Languages: English, French, Spanish, German.

## CAREER

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**University of Houston**

2024 +

Assistant professor

**University of Zürich**

2023-2024

Postdoctoral researcher

**Rutgers University**

2020-2023

Hill assistant professor

## EDUCATION

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**University of Bristol**

2016-2020

PhD in Mathematics

Advisors: Jens Marklof & Bálint Tóth

**University of Texas at Austin**

2012-2016

B.S Mathematics

Dean's Scholars Honors Program

## PUBLICATIONS

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### Preprints:

28. *Spacing statistics for a point scatterer on the cubic three-torus* (2026).
27. *Spherical density for real semisimple groups*, arXiv:2610.00298 (2026).
26. *Loss of memory in the periodic Ehrenfest model with small polyhedral scatterers* with J. Marklof, arXiv:2609.04371 (2026).
25. *Poissonian pair correlation for the three-particle Sutherland model*, arXiv:2609.03112 (2026).
24. *Improved sup-norm bounds for locally symmetric spaces*, arXiv:2608.21580 (2026).
23. *Poissonian correlations of  $\alpha n^d \bmod 1$*  with N. Rome and N. Technau, arXiv:2605.06974 (2026).
22. *The Gauss circle problem for Penrose tilings* with A. Haynes, arXiv:2512.21444 (2025).
21. *Average variance bounds for integer points on the sphere*, arXiv:2402.12822 (2024).

### Papers:

20. *Exceptional eigenvalue density for thin groups*,  
**Int. Math. Res. Not. (IMRN)**, **2026**(18), 208 (2026).
19. *Sign changes along geodesics of modular forms*  
with D. Kelmer and A. Kontorovich (2024) [Accepted: **J. Théor. Nombres Bordeaux**].
18. *Counting in lattice orbits*  
with A. Kontorovich (2024) [Accepted: **Bull. London Math. Soc.**].
17. *Diffusion of the random Lorentz process in a magnetic field*  
with B. Tóth, **J. Math. Phys.**, **66**(11), (2025) (Editor's pick).
16. *Polyhedral bounds on the joint spectrum and temperedness of locally symmetric spaces*  
with T. Weich and L. Wolf (2024) [Accepted: **Duke Math. J.**].
15. *An abstract spectral approach to horospherical equidistribution*,  
**Nonlinearity**, **38**(10), 105014, (2025).
14. *Hyperbolic lattice point counting in unbounded rank*  
with V. Blomer, **J. Reine Angew. Math.**, **2024**(812), 257-274, (2024).
13. *Norm bounds on Eisenstein series*  
with D. Kelmer and A. Kontorovich, **Int. J. Number Theory**, **20** (08), 2083-2098 (2024).
12. *These Numbers Look Random but Aren't, Mathematicians Prove*  
**Scientific American** (2024) (also featured in **Spektrum der Wissenschaft**).
11. *m-Point Correlations of the Fractional Parts of  $\alpha n^\theta$*   
with N. Technau, **Amer. J. Math.**, **148**(2), 569-597 (2026).
10. *Full poissonian local statistics of slowly growing sequences*  
with N. Technau, **Compos. Math.**, **161**(1), 148-180, (2025).
9. *Effective counting in sphere packings*  
with A. Kontorovich, **J. of the Assoc. Math. Res.**, **2**, 15-52 (2024).
8. *Sarnak's spectral gap question*  
with D. Kelmer and A. Kontorovich, **J. Anal. Math.** **151**, 171-179 (2023) (special volume for P. Sarnak).
7. *Pair correlation of the fractional parts of  $\alpha n^\theta$*   
with A. Sourmelidis and N. Technau, **J. of the Eur. Math. Soc. (JEMS)**, **27**(10), 4069-4082 (2024).
6. *Long-range correlations of sequences modulo 1*,  
**J. of Number Theory**, **234**, 333-348 (2022).
5. *Farey sequences for thin groups*  
**Int. Math. Res. Not. (IMRN)**, **15**, 11642-11689 (2022).
4. *Invariance principle for the random wind-tree process*  
with B. Tóth, **Ann. Henri Poincaré**, **22**(10), 3357-3389 (2021).
3. *Invariance principle for the random Lorentz gas – beyond the Boltzmann-Grad limit*  
with B. Tóth, **Comm. in Math. Phys.**, **379**, 589-632, (2020).
2. *Directions in orbits of geometrically finite hyperbolic subgroups*,  
**Math. Proc. of the Cambridge Philos. Soc.** **171**(2), 277-316 (2021).

1. *A microscopic approach to nonlinear Reaction-Diffusion: the case of morphogen gradient formation*  
with J.P. Boon, J.F. Lutsko, **Phys. Rev. E** 85:021126 (2016).

### PhD Thesis:

Statistical Properties of Dynamical Systems: From Statistical Mechanics to Hyperbolic Geometry, 2020 (Bristol University). [Nominated for doctoral dissertation prize]

### Conference Proceedings:

1. *Invariance principle for random Lorentz gas in the Boltzmann-Grad Limit*, Oberwolfach Report 10/2019 p. 33-35 (2019)
2. *Invariance principle for random Lorentz gas — Beyond the Boltzmann-Grad Limit*, Oberwolfach Report 42/2019 p. 12-15 (2019)

## FUNDING AND AWARDS

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|                                                                      |      |
|----------------------------------------------------------------------|------|
| Simons Travel Grant (5 years, \$35,000)                              | 2025 |
| Nominated for doctoral dissertation prize for outstanding excellence | 2020 |
| William's Scholar Honored Graduate                                   | 2016 |

## TEACHING

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### Instructor:

|                                                   |             |
|---------------------------------------------------|-------------|
| Houston: Math 6322 Functions of Complex Variables | Fall 2026   |
| Houston: Math 2451 Accelerated Calculus           | Spring 2026 |
| Houston: Math 3338 Probability                    | Fall 2025   |
| Houston: Math 2451 Accelerated Calculus           | Spring 2025 |
| Houston: Math 2450 Accelerated Calculus           | Fall 2024   |
| Rutgers: Math 152 Calculus II (Math/Physics)      | Spring 2023 |
| Rutgers: Math 152 Calculus II (Math/Physics)      | Spring 2023 |
| Rutgers: Math 356 Theory of Numbers               | Fall 2022   |
| Rutgers: Math 437 History of Mathematics          | Spring 2022 |
| Rutgers: Math 152 Calculus II (Math/Physics)      | Spring 2022 |
| Rutgers: Math 356 Theory of Numbers               | Fall 2021   |
| Rutgers: Math 356 Theory of Numbers               | Spring 2021 |
| Rutgers: Math 152 Calculus II (Math/Physics)      | Spring 2021 |
| Rutgers: Math 152 Calculus II (Math/Physics)      | Fall 2020   |

### Teaching Assistant:

Acted as TA for approximately 11 different courses from 2013-2024

## POST-DOCS

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|               |       |
|---------------|-------|
| Kavita Dhanda | 2026+ |
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## GRADUATE STUDENTS

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|              |       |
|--------------|-------|
| Anish Ray    | 2025+ |
| Ziyue Song   | 2026+ |
| Thomas Zhang | 2026+ |

## UNDERGRADUATE MENTORSHIP

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Arin Morantes (Undergraduate thesis advisor)

Summer 2026+

## ORGANIZATIONAL DUTIES

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Linfoot number theory seminar (Bristol)

Fall 2019 - Spring 2020

Rutgers number theory seminar

Fall 2021 - Spring 2023

Houston dynamics seminar

Fall 2024 +

Rice-UH Dynamics Day (once per semester)

Spring 2026 +

## ACADEMIC INVITATIONS

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1 week - Budapest University of Technology (BME)

2017

2 weeks - University of Bonn

2023

1 week - UW Madison

2026

## SEMINAR TALKS

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University of Bristol

2017

Budapest University of Technology

2017

University of Bristol

2017

University of Warwick

2018

University of Bristol

2018

University of Bristol

2018

Univ. Libre de Bruxelles

2018

University of Bristol

2018

Rutgers University

2019

University of Texas at Austin

2019

University of Houston

2019

University of Exeter

2019

Manchester University

2019

University of Bristol

2019

University of Loughborough

2020

University of Oklahoma

2020

Yeshiva University

2020

Rutgers University

2021

Tata Institute of Fundamental Research

2021

New England Dynamics and Number Theory Seminar

2021

(recording: <https://www.youtube.com/watch?v=EODTepggUuU>)

Hong Kong University

2022

University of Illinois Urbana-Champaign

2022

Stony Brook **Colloquium**

2022

(recording: <https://www.youtube.com/watch?v=SOHhKdno2jA>)

Rutgers University

2022

New England Dynamics and Number Theory Seminar

2022

(recording: <https://nednt.wescreates.wesleyan.edu/fall-2022-schedule/>)

University of Maryland

2022

Philadelphia Area Number Theory Seminar (Bryn Mawr)

2023

Rutgers University

2023

University of Bonn

2023

Paderborn University

2023

University of Zurich

2023

EPFL

2024

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|---------------------------------------------------------------------------------------------------------------------|------|
| University of Bristol                                                                                               | 2024 |
| Manchester University ( $\times 2$ )                                                                                | 2024 |
| University of Houston <b>Colloquium</b>                                                                             | 2024 |
| Rice University                                                                                                     | 2024 |
| One World Probability Seminar                                                                                       | 2024 |
| Yale University                                                                                                     | 2024 |
| University of Houston                                                                                               | 2025 |
| IST Austria - University of Graz Joint Seminar                                                                      | 2025 |
| UW Madison                                                                                                          | 2026 |
| International Seminar on Automorphic Forms                                                                          | 2026 |
| (recording: <a href="https://www.youtube.com/watch?v=PuWv7AB5Tgg">https://www.youtube.com/watch?v=PuWv7AB5Tgg</a> ) |      |
| Rice University                                                                                                     | 2026 |

## CONFERENCE TALKS

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|                                                                                       |      |
|---------------------------------------------------------------------------------------|------|
| MINGLE post-graduate event - University of Bristol, UK                                | 2017 |
| Dynamics Days Europe (Billiards Minisymposium) - University of Loughborough, UK       | 2018 |
| Probability and NonLocal PDEs - University of Swansea, UK                             | 2018 |
| Mini-Workshop: Lorentz Gas Dynamics - M. F. Oberwolfach, Germany                      | 2019 |
| Large Scale Stochastic Dynamics - M. F. Oberwolfach, Germany                          | 2019 |
| MINGLE post-graduate event - University of Bristol, UK                                | 2019 |
| CRC Workshop "Geometry and analysis of locally symmetric spaces" - Paderborn, Germany | 2024 |
| Stochastics and Influences - Erdős Center, Budapest, Hungary                          | 2025 |
| Rice-UH Dynamics Day                                                                  | 2026 |
| BIRS Workshop, "Arithmetic, L-functions, and Pseudorandomness" - Banff, Canada        | 2026 |

## OUTREACH

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|                                                                                                                 |      |
|-----------------------------------------------------------------------------------------------------------------|------|
| Volunteer Chemistry Tutor – Garza high school (Austin, Texas)                                                   | 2015 |
| Help with Bristol maths day                                                                                     | 2018 |
| Participant in "Skype a Scientist"                                                                              | 2021 |
| Speaker at Chicago public schools teen mentoring program STEM event                                             | 2021 |
| Contributor to EMS magazine problems section                                                                    | 2022 |
| Help with Museum of mathematics' polyplane project                                                              | 2023 |
| Contributor to Scientific American [link] (this article was also featured in <b>Spektrum der Wissenschaft</b> ) | 2024 |
| Interview on ABC 13 Houston (Houston area television news program)                                              | 2025 |

## REFEREEING/REVIEWING/SERVICE

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AMS Mathematical Reviews/MathSciNet [12 reviews] • Advances in Mathematics • American Journal of Mathematics • Annales Henri Poincaré • Chaos, Solitons and Fractals • Communications of the AMS • Compositio Mathematica • Ergodic Theory and Dynamical Systems • Indagationes Mathematicae • International Mathematics Research Notices • Journal of the European Mathematical Society • Journal of the London Mathematical Society • Journal of Mathematical Physics • Journal of Number Theory • Journal of Statistical Physics • Mathematical Proceedings of the Cambridge Philosophical Society • Nonlinearity • Random Structures and Algorithms • Research in Number Theory • Stochastic Processes and their Applications